

E.3 / PAPER 2 / QUESTION 4

3 November 2022 / TZ0 / HL Paper 2 / original Q6 / 8822-6502

4. (a) Outline, by reference to nuclear binding energy, why the mass of a nucleus is less than the sum of the masses of its constituent nucleons. [2]

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Polonium-210 (Po-210) decays by alpha emission into lead-206 (Pb-206).

The following data are available.

Nuclear mass of Po-210 = 209.93676 u

Nuclear mass of Pb-206 = 205.92945 u

Mass of the alpha particle = 4.00151 u

- (b) (i) Calculate, in MeV, the energy released in this decay. [2]

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- (ii) The polonium nucleus was stationary before the decay.

Show, by reference to the momentum of the particles, that the kinetic energy of the alpha particle is much greater than the kinetic energy of the lead nucleus. [3]

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(Question 4 continued)

- (iii) In the decay of polonium–210, alpha emissions can be accompanied by the emissions of gamma photons, all of the same wavelength of $1.54 \times 10^{-12} \text{ m}$.

Discuss how this observation provides evidence for discrete nuclear energy levels. [3]

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- (c) A sample contains 5.0 g of pure polonium-210. The decay constant of polonium-210 is $5.8 \times 10^{-8} \text{ s}^{-1}$. Lead-206 is stable.

Calculate the mass of lead-206 present in the sample after one year. [3]

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